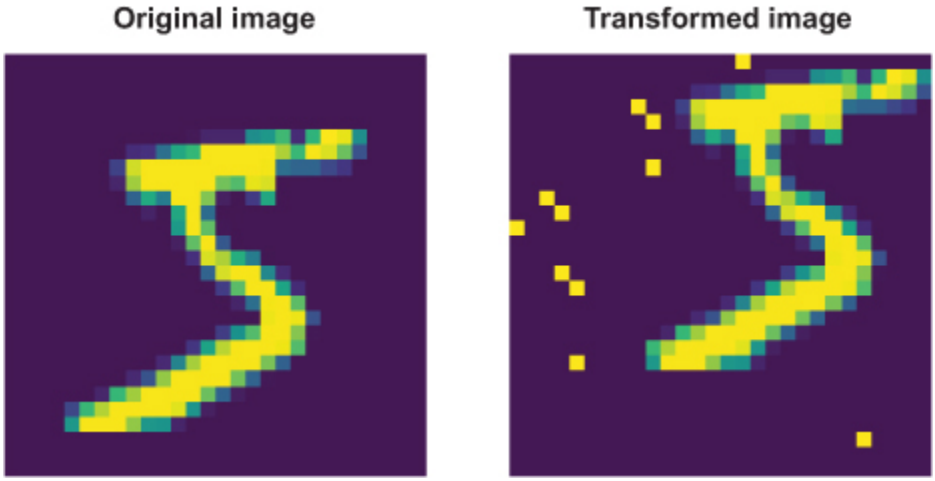
Figure 10.14. Left: Original MNIST digit for the number “5”. Right: Transformed version that is translated to the top right and with random noise sprinkled in.



10.3.2. The relational module

Now we can dive into the relational neural network itself. Until now, all of the projects in this book were designed to be compelling enough to illustrate an important concept but simple enough to be able to run on a modern laptop without needing a GPU. The computational demands of the self-attention module, however, are significantly greater than any of the other models we have built so far in the book. You can still try running this model on your laptop, but it will be significantly faster if you have a CUDA-enabled GPU. If you don’t have a GPU, you can easily launch a cloud-based Jupyter Notebook using Amazon SageMaker, Google Cloud, or Google Colab (which is free as of this writing).

Note

The code we show in this book will not include the necessary (but very minor) modifications necessary to run on a GPU. Please refer to this book’s GitHub page at http://mng.bz/JzKp to see how to enable the code to run on a GPU, or consult the PyTorch documentation at https://pytorch.org/docs/stable/notes/cuda.html.
